

CHAPTER 2

SOFTWARE REQUIREMENTS AND DESIGN SPECIFICATION

APPLICATION OVERVIEW

TransiTrack is a comprehensive bus transportation management system designed to streamline and modernize public bus operations through integrated digital solutions. The platform serves as a centralized solution that connects four key stakeholders in the public transportation ecosystem: commuters, bus drivers, bus operators, and terminal managers, each with specialized interfaces tailored to their unique operational needs and responsibilities.

For commuters, TransiTrack provides a user-friendly mobile application built on Ionic framework where passengers can view real-time bus locations through Mapbox integration, track arrival times with coordinate-based precision, check route schedules, and plan their journeys with integrated fare calculations. The app displays live bus positions on interactive maps with Firebase real-time synchronization, allowing commuters to make informed decisions about their travel plans. Additionally, users can access terminal bay information, route endpoints with precise coordinates, and receive notifications about service disruptions or schedule changes, ensuring they stay informed throughout their journey with accurate boarding location details.

Bus drivers benefit from a dedicated dashboard interface that serves as their digital companion during daily operations. The driver application provides access to assigned route information with coordinate-based mapping, schedule details managed through the Laravel backend, and real-time navigation assistance using Mapbox visualization. Drivers can view their routes with precise endpoint coordinates, update their operational status, and access schedule information that reflects real-time updates from operator modifications. The system tracks driver performance metrics through database logging, including punctuality and route adherence, helping drivers maintain professional standards while ensuring passenger safety and service reliability.

The bus operator web dashboard serves as the operational nerve center, providing comprehensive management tools for fleet oversight and schedule coordination. Operators can monitor all active buses in real-time, manage driver

assignments through role-based authentication, create and modify route schedules with coordinate input, and oversee daily operations from designated terminals with MySQL database backend. The dashboard displays critical information such as bus locations through coordinate tracking, schedule management with date-based filtering, and terminal bay assignments. Operators can generate detailed reports on fleet performance, route efficiency based on coordinate accuracy, and operational analysis through database queries. The system provides a visual terminal layout showing all available bays, their current occupancy status, and scheduled assignments for passenger boarding areas, making terminal space management intuitive and efficient.

Terminal managers have access to a specialized administrative interface that focuses on terminal operations and inter-terminal coordination through secure authentication middleware. This system component manages passenger flow through bay assignment controls, terminal capacity optimization, oversees bus schedules with cross-terminal visibility, and coordinates with multiple terminal facilities through centralized database management. Terminal managers can monitor arrivals and departures through real-time tracking, manage terminal resources including parking and boarding bay allocations, oversee operational protocols, and ensure compliance with transportation regulations. The system facilitates communication between different terminals through shared database access, enabling coordinated responses to service disruptions or operational changes.

TransiTrack addresses critical challenges in terminal-based public transportation systems, including unpredictable schedules through database-driven scheduling, lack of real-time coordinate information, inefficient route management through precise mapping, and poor communication between stakeholders via role-based interfaces. By providing transparency through coordinate tracking, accountability through performance monitoring, and efficiency tools through optimized bay management, the platform aims to enhance the overall quality of public bus transportation, reduce waiting times through accurate arrival predictions, optimize fleet utilization through data-driven scheduling, and create a more reliable and sustainable transportation network that serves operational needs through integrated terminal management systems."

USE CASE DIAGRAM

Use case diagram illustrate how actors relate to each other.

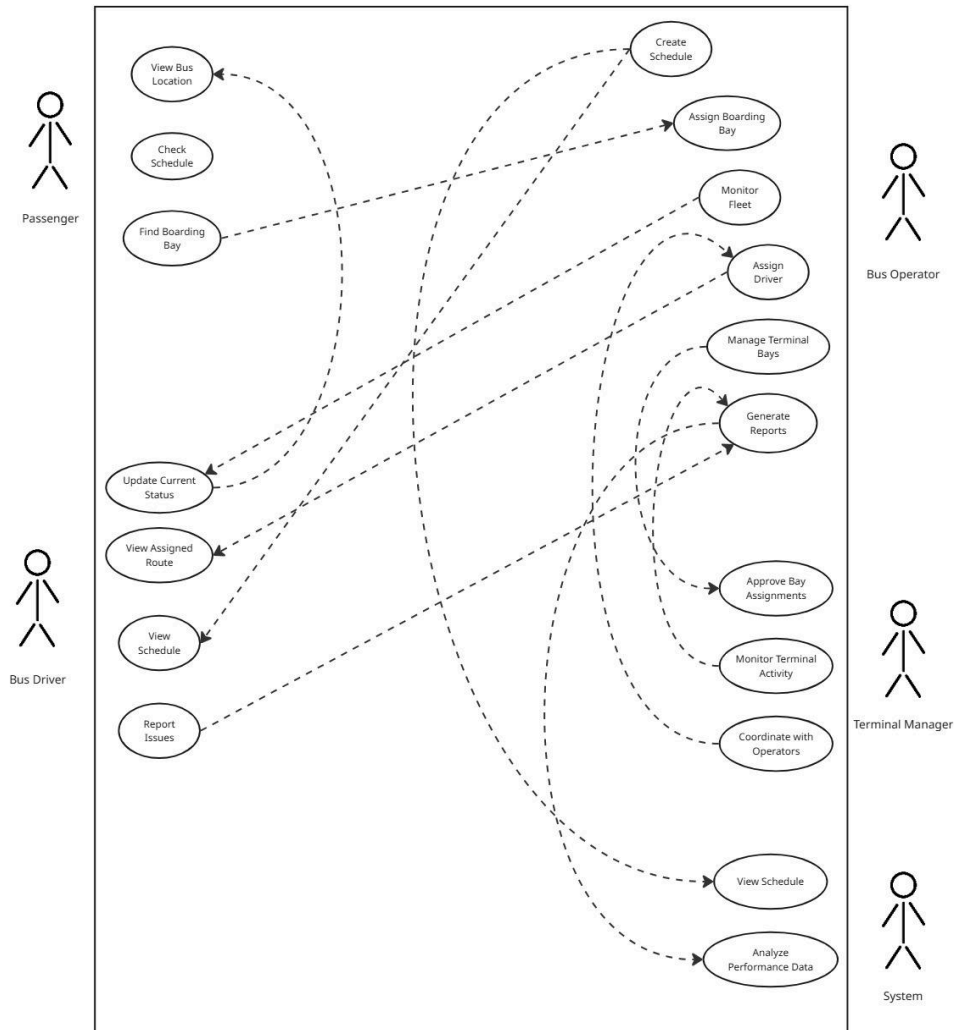


Figure 1: TransiTrack Bus Transportation Management System

Figure 1 illustrates the comprehensive use case relationships within the TransiTrack Bus Transportation Management System. It demonstrates how each stakeholder; Bus Operator, Bus Driver, Commuter, Terminal Manager, and System interacts through interconnected functionalities including schedule management, fleet monitoring, route assignments, terminal bay coordination, and real-time tracking. This diagram establishes the foundational framework of system interactions and defines the operational responsibilities across all user roles within the integrated transportation management platform.

ARCHITECTURAL DIAGRAM

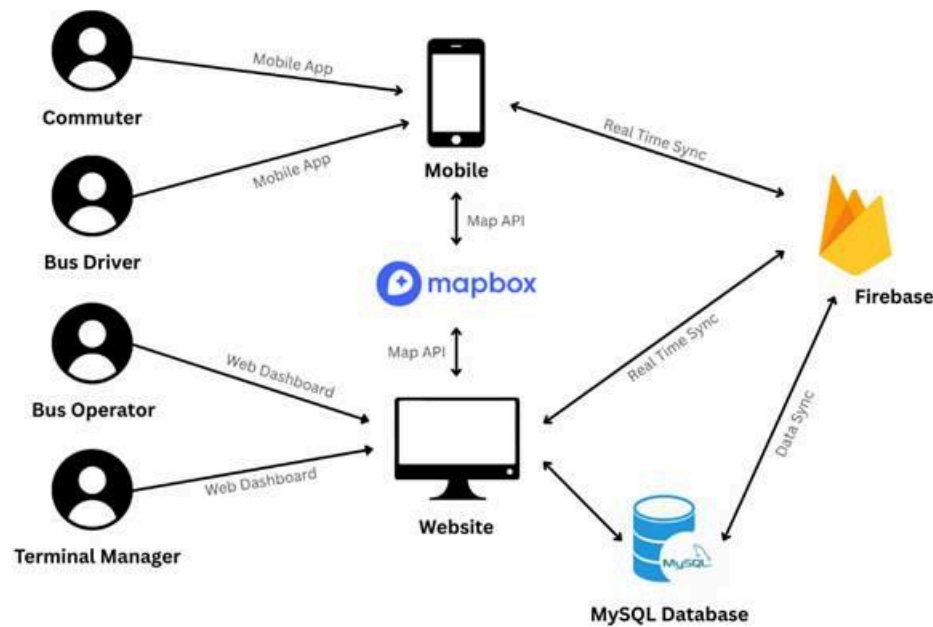


Figure 2: Architectural Diagram

Figure 2 illustrates the comprehensive architectural framework of the TransiTrack Bus Transportation Management System, showcasing the interconnected flow between users, interfaces, and backend services. The architecture employs a dual-interface approach where Commuters and Bus Drivers access real-time transportation data through the TransiTrack Mobile Application built on Ionic framework, while Bus Operators and Terminal Managers utilize the Laravel-powered Bus Operator Web Dashboard for administrative functions. Both interfaces integrate with Mapbox for GPS tracking and route visualization, ensuring accurate location services across all user interactions.

The system's data management operates through a synchronized multi-database architecture where the MySQL Database serves as the primary storage for operational data including schedules, routes, and user management, accessible through Laravel API connections from the web dashboard. Firebase provides real-time synchronization capabilities, bridging mobile application requirements with backend database operations through bidirectional data flow. This ensures that schedule updates, driver assignments, and route modifications made by operators instantly propagate to drivers and commuters on their mobile devices.

The architectural design creates a seamless operational ecosystem where administrative actions flow from web-based management tools to real-time mobile updates, while field data from drivers and GPS tracking feeds back into the central system for comprehensive fleet monitoring. This integrated approach supports efficient transportation management from strategic planning through the web dashboard to real-time execution via mobile applications, maintaining data consistency and operational coordination across all system components.

USE CASE NARRATIVES

Use Case 01: User Login and Authentication

Use Case	User Login and Authentication
Actors	Terminal Manager, Bus Operator, Bus Driver, Commuter
Pre-condition	- User has a valid registered account in the TransiTrack system - User has access to the appropriate platform (web dashboard for operators/drivers, mobile app for commuters) - System database is operational and accessible - Network connectivity is available

Post-Condition	- User is successfully authenticated and logged into the system - User session is established with appropriate role-based permissions - User is redirected to their respective dashboard/interface - Authentication token is generated and stored for session management - User activity is logged in the system
Flow of Events	
Actor action	System Response

1. User accesses the TransiTrack login interface (web dashboard or mobile app) 2. User enters their registered email address in the email field 3. User enters their password in the password field 4. User selects their assigned terminal from the dropdown menu (applicable for operators and drivers) 5. User optionally checks the "Remember Me" checkbox for persistent login 6. User clicks the "Login" button to submit authentication credentials 7. User waits for authentication verification	1. System displays the login form with email and password fields, along with "Remember Me" option and terminal selection (for operators/drivers) 2. System validates the email format and highlights the field as valid input 3. System masks the password input for security and enables the login button 4. System displays the selected terminal and validates terminal assignment permissions 5. System acknowledges the remember preference setting 6. System validates credentials against the database, verifies terminal assignment, checks user role permissions, and processes authentication request 7. System generates authentication token, establishes user session, logs login activity, and redirects user to their role-specific dashboard (Operator Panel, Driver Dashboard, or Commuter Interface)
Alternative Scenarios	

Actor Action	System Response
1a. User enters incorrect email or password combination 2a. Operator or driver attempts to access from unauthorized terminal 3a. User submits login form during network interruption	1a. System displays error message "Invalid credentials. Please check your email and password." and returns to login form with fields cleared 2a. System displays error message "Access denied. You are not authorized to access this terminal." and prevents login 3a. System displays connection error message "Unable to connect. Please check your internet connection and try again."

Table 1. User Login and Authentication

Use Case 02: Schedule Creation and Management

Use Case	Schedule Creation and Management
Actors	Bus Operator
Pre-condition	The operator is logged in and has access to schedule management features.
Post-Condition	A new schedule is created and assigned to drivers with route coordination.

Flow of Events	
Actor action	System Response
1. The operator clicks "Create New Schedule" button on the dashboard. 2. The operator selects route, driver, bus, and departure time from dropdowns. 3. The operator assigns terminal bay for passenger boarding. 4. The operator saves the schedule and confirms the assignment.	1. The system displays the schedule creation form with available options. 2. The system validates selections and checks for conflicts with existing schedules. 3. The system stores the schedule in the database and updates bay assignments. 4. The system sends notifications to assigned drivers and updates real-time displays.
Alternative Scenarios	
Actor Action	System Response
1a. The operator tries to assign a driver who's already scheduled.	1a. The system displays "Driver already assigned for this time" error.
2a. The operator selects a bay that's already occupied.	2a. The system shows available alternative bays and suggests reassignment.
3a. The operator cancels the schedule creation process.	3a. The system discards entered information and returns to the dashboard.

Table 2. Schedule Creation and Management

Use Case 03: Fleet Monitoring and Driver Management

Use Case	Fleet Monitoring and Driver Management
Actors	Bus Operator
Pre-condition	The operator is logged in and buses are active in the system.
Post-Condition	The operator successfully monitors fleet status and manages driver assignments.
Flow of Events	
Actor action	System Response
1. The operator accesses the "Fleet Monitor" section from the dashboard. 2. The operator views real-time bus locations and driver status updates. 3. The operator assigns or reassigns drivers to specific routes as needed. 4. The operator generates performance reports for analysis.	1. The system displays a real-time map with all active buses and their coordinates. 2. The system shows current driver status, route progress, and any reported issues. 3. The system updates assignments in the database and notifies affected drivers. 4. The system compiles data and generates comprehensive fleet performance reports.
Alternative Scenarios	

Actor Action	System Response
1a. The operator tries to reassign a driver who's currently on route.	1a. The system warns about active route status and requires confirmation.
2a. The operator requests reports for a date range with no data.	2a. The system displays "No data available for selected period" message.
3a. The operator encounters a driver who's reported an emergency.	3a. The system highlights the emergency status and provides emergency contact options.

Table 3. Fleet Monitoring and Driver Management

Use Case 04: Terminal Bay Management

Use Case	Terminal Bay Management
Actors	Bus Operator
Pre-condition	The operator is logged in and has access to terminal bay management features.
Post-Condition	Terminal bays are properly assigned and managed for optimal passenger flow.
Flow of Events	
Actor action	System Response

1. The operator accesses the "Terminal Bay Management" section from the dashboard. 2. The operator views the visual terminal layout showing all bay statuses. 3. The operator assigns specific bays for parking and passenger boarding areas. 4. The operator updates bay assignments based on route schedules and passenger volume.	1. The system displays a visual terminal map with all available bays and their current status. 2. The system shows real-time occupancy, scheduled assignments, and bay availability. 3. The system validates bay assignments and checks for scheduling conflicts. 4. The system updates bay information and notifies relevant personnel of changes.
Alternative Scenarios	
Actor Action	System Response
1a. The operator tries to assign an already occupied bay. 2a. The operator attempts to assign a bay that's under maintenance. 3a. The operator needs to reassign bays due to emergency situations.	1a. The system displays "Bay currently occupied" and suggests alternative bays. 2a. The system shows "Bay unavailable for maintenance" and blocks assignment. 3a. The system allows emergency reassignment and sends priority notifications.

Table 4. Terminal Bay Management

Use Case 05: Route Creation and Coordinate Management

Use Case	Route Creation and Coordinate Management
Actors	Bus Operator
Pre-condition	The operator is logged in and has access to route management features.
Post-Condition	New routes are created with precise coordinates for driver navigation.
Flow of Events	
Actor action	System Response
1. The operator clicks "Create New Route" and enters route details. 2. The operator sets start and end coordinates using the map interface. 3. The operator adds intermediate stops with their respective coordinates. 4. The operator saves the route and makes it available for schedule assignment.	1. The system displays the route creation form with integrated Mapbox interface. 2. The system validates coordinate accuracy and calculates route distance. 3. The system stores route data with coordinate information in the database. 4. The system makes the route available in schedule creation and driver dashboards.
Alternative Scenarios	
Actor Action	System Response

1a. The operator enters invalid coordinates.	1a. The system displays "Invalid coordinates" error and requests correction.
2a. The operator creates a route that conflicts with existing routes.	2a. The system warns about route conflicts and suggests modifications.
3a. The operator cancels route creation mid-process.	3a. The system discards unsaved route data and returns to route management.

Table 5. Route Creation and Coordinate Management

Use Case 06: Driver Assignment and Schedule Management

Use Case	Driver Assignment and Schedule Management
Actors	Bus Operator
Pre-condition	The operator is logged in and has access to driver management features.
Post-Condition	Drivers are properly assigned to schedules with appropriate routes and buses.
Flow of Events	
Actor action	System Response

1. The operator accesses the "Drivers" section from the dashboard cards. 2. The operator views the list of available drivers and their current status. 3. The operator selects a driver and views their current schedule information. 4. The operator assigns drivers to specific routes and schedules through the schedule creation panel.	1. The system displays all drivers with their roles, status (active/inactive), and current assignments. 2. The system shows driver availability, current routes, and terminal assignments (North Terminal). 3. The system provides driver schedule details including current route coordinates and daily schedules. 4. The system validates driver assignments, checks for conflicts, and updates the database with new assignments.
Alternative Scenarios	
Actor Action	System Response
1a. The operator tries to assign a driver who's already scheduled for the same time. 2a. The operator needs to view a driver's detailed profile and statistics. 3a. The operator wants to reassign a driver from one route to another.	1a. The system displays "Driver already scheduled" error and suggests alternative time slots. 2a. The system shows driver profile with mock statistics (trips completed, on-time rate, total distance, fuel efficiency). 3a. The system allows reassignment and updates both the old and new schedule assignments.

Table 6. Driver Assignment and Schedule Management

Use Case 07: Bus Fleet Management

Use Case	Bus Fleet Management
Actors	Bus Operator
Pre-condition	The operator is logged in and has access to bus management features.
Post-Condition	Bus fleet information is properly maintained and buses are assigned to active service.
Flow of Events	
Actor action	System Response
1. The operator clicks on the "Buses in Operation" card from the dashboard. 2. The operator views the list of all buses with their current status and information. 3. The operator adds new buses or updates existing bus information (license plate, capacity, status). 4. The operator assigns buses to specific routes and schedules.	1. The system displays the buses panel with all registered buses. 2. The system shows bus details including ID, license plate, capacity, and operational status. 3. The system validates bus information and updates the database with changes. 4. The system makes buses available for schedule assignment and route planning.
Alternative Scenarios	

Actor Action	System Response
1a. The operator tries to assign a bus that's already in use. 2a. The operator needs to mark a bus as out of service for maintenance. 3a. The operator wants to view bus utilization statistics.	1a. The system displays "Bus already assigned" and shows current assignment details. 2a. The system updates bus status and removes it from available assignments. 3a. The system provides bus usage reports and operational statistics.

Table 7. Bus Fleet Management

Use Case 08: User Profile and System Settings Management

Use Case	User Profile and System Settings Management
Actors	Bus Operators
Pre-condition	The operator is logged in and has access to profile settings.
Post-Condition	User profile information is updated and system preferences are configured.
Flow of Events	
Actor action	System Response

1. The operator accesses the profile section from the navigation menu. 2. The operator reviews and updates personal information and contact details. 3. The operator configures system preferences and notification settings. 4. The operator saves changes and confirms the updates.	1. The system displays the current profile information and available settings. 2. The system validates the updated information for completeness and accuracy. 3. The system applies the new settings and updates user preferences. 4. The system confirms successful updates and applies changes to the user session.
Alternative Scenarios	
Actor Action	System Response
1a. The operator tries to update with invalid email format. 2a. The operator wants to change password. 3a. The operator needs to update terminal assignment.	1a. The system displays validation errors and requests correct format. 2a. The system requires current password verification before allowing change. 3a. The system validates terminal permissions and updates assignment accordingly.

Table 8. User Profile and System Settings Management

Use Case 09: System Notifications and Alerts Management

Use Case	System Notifications and Alerts Management
Actors	Bus Operator
Pre-condition	The operator is logged in and has access to notification features.
Post-Condition	Important system notifications are reviewed and appropriate actions are taken.
Flow of Events	
Actor action	System Response
1. The operator accesses the notifications panel from the dashboard. 2. The operator reviews system alerts, driver reports, and operational updates. 3. The operator acknowledges important notifications and takes necessary actions. 4. The operator configures notification preferences for different types of alerts.	1. The system displays all pending notifications categorized by type and priority. 2. The system shows notification details including timestamp, source, and required actions. 3. The system marks notifications as read and logs operator responses. 4. The system updates notification settings and applies preferences for future alerts.
Alternative Scenarios	

Actor Action	System Response
1a. The operator receives emergency notifications requiring immediate action. 2a. The operator wants to filter notifications by date or type. 3a. The operator needs to forward notifications to other personnel.	1a. The system highlights emergency notifications and provides quick action buttons. 2a. The system applies filters and displays relevant notifications only. 3a. The system allows notification forwarding and logs the action for audit purposes.

Table 9. System Notifications and Alerts Management

Use Case 10: Dashboard Overview and Quick Actions

Use Case	Dashboard Overview and Quick Actions
Actors	Bus Operator
Pre-condition	The operator is logged in and accessing the main dashboard.
Post-Condition	The operator has a comprehensive view of operations and can perform quick actions.
Flow of Events	
Actor action	System Response

1. The operator logs in and views the main operator dashboard. 2. The operator reviews real-time statistics and operational metrics. 3. The operator uses quick action buttons for common tasks (Quick Schedule, Refresh). 4. The operator navigates to specific modules using dashboard cards.	1. The system displays a dashboard with current statistics (active routes, buses, drivers, issues). 2. The system shows recent schedules, today's statistics, and performance metrics. 3. The system provides quick access to frequently used functions and real-time data refresh. 4. The system enables seamless navigation to detailed management panels.
Alternative Scenarios	
Actor Action	System Response
1a. The operator needs to view historical performance data. 2a. The operator wants to customize the dashboard layout. 3a. The operator requires access to advanced analytics.	1a. The system provides access to historical reports and trend analysis. 2a. The system allows dashboard customization within available options. 3a. The system redirects to detailed analytics and reporting sections.

Table 10. Dashboard Overview and Quick Actions

Use Case 11: Driver Authentication and Profile Access

Use Case	Driver Authentication and Profile Access
Actors	Bus Driver
Pre-condition	The driver has valid credentials and access to the system.
Post-Condition	The driver is successfully authenticated and can access their profile and schedule information.
Flow of Events	
Actor action	System Response
1. The driver logs into the system using their credentials. 2. The driver accesses their profile to view personal information and license details. 3. The driver reviews their profile status (active, inactive, on-leave). 4. The driver can view their contact information and emergency contact details.	1. The system validates credentials and creates a driver session. 2. The system displays a driver profile with personal information, license expiry, and photo. 3. The system shows current status and any restrictions or notifications. 4. The system provides access to profile management and schedule viewing features.
Alternative Scenarios	
Actor Action	System Response

1a. The driver enters incorrect login credentials.	1a. The system displays an "Invalid credentials" error message.
2a. The driver's license has expired.	2a. The system displays license expiry warning and restricts schedule access.
3a. The driver's status is set to inactive or on-leave.	3a. The system shows status notification and limited access to features.

Table 11. Driver Authentication and Profile Access

Use Case 12: Schedule Viewing and Route Information

Use Case	Schedule Viewing and Route Information
Actors	Bus Driver
Pre-condition	The driver is logged in and has been assigned schedules.
Post-Condition	The driver can view their current and upcoming schedules with route details.
Flow of Events	
Actor action	System Response

1. The driver accesses the driver dashboard to view assigned schedules. 2. The driver selects their name from the driver dropdown to view their specific information. 3. The driver reviews today's schedule and upcoming assignments. 4. The driver checks route details, bus assignments, and timing information.	1. The system displays the driver dashboard with a schedule overview. 2. The system loads driver-specific schedule data and route assignments. 3. The system shows current route, bus information, capacity, start/end times, and status. 4. The system provides a detailed schedule list with route names, times, and assignment status.
Alternative Scenarios	
Actor Action	System Response
1a. The driver has no scheduled assignments for the day. 2a. The driver needs to view historical schedule data. 3a. The driver wants to check for schedule changes or updates.	1a. The system displays "No schedules assigned for today" message. 2a. The system provides access to historical schedule records with filtering options. 3a. The system shows recent schedule modifications and notifications about changes.

Table 12. Schedule Viewing and Route Information

Use Case 13: Route Navigation and Map Viewing

Use Case	Route Navigation and Map Viewing
Actors	Bus Driver
Pre-condition	The driver is logged in and has an active route assignment.
Post-Condition	The driver can view their route on an interactive map with coordinate-based navigation.
Flow of Events	
Actor action	System Response
1. The driver accesses the route map section from the dashboard. 2. The driver views their assigned route with start and end coordinates. 3. The driver uses the interactive map to understand the route path and key locations. 4. The driver references the map during their journey for navigation assistance.	1. The system displays an interactive Mapbox-powered map interface. 2. The system plots the route using stored start and end coordinates from the database. 3. The system shows route markers, terminal locations, and key waypoints on the map. 4. The system provides real-time route visualization with coordinate-based navigation support.
Alternative Scenarios	

Actor Action	System Response
1a. The driver's assigned route has missing or incorrect coordinates. 2a. The driver needs to report route issues or obstacles. 3a. The driver requires alternative route suggestions.	1a. The system displays default coordinates and notifies administrators of missing data. 2a. The system provides incident reporting functionality to flag route problems. 3a. The system suggests alternative routes based on available coordinate data.

Table 13. Route Navigation and Map Viewing

Use Case 14: Bus Assignment and Vehicle Information

Use Case	Bus Assignment and Vehicle Information
Actors	Bus Driver
Pre-condition	The driver is logged in and has been assigned to a specific bus.
Post-Condition	The driver can view their assigned bus details and capacity information.
Flow of Events	
Actor action	System Response

1. The driver checks their current bus assignment from the dashboard. 2. The driver reviews bus specifications including license plate and capacity. 3. The driver verifies bus status and operational readiness. 4. The driver acknowledges the bus assignment for their scheduled route.	1. The system displays currently assigned bus information in the route details section. 2. The system shows the bus license plate, passenger capacity, and current operational status. 3. The system provides bus maintenance status and any operational restrictions. 4. The system confirms bus assignment and links it to the driver's route schedule.
Alternative Scenarios	
Actor Action	System Response
1a. The driver is assigned to a bus that's under maintenance.	1a. The system displays maintenance status and prevents route assignment.
2a. The driver needs to report bus mechanical issues.	2a. The system provides issue reporting functionality and alerts maintenance staff.
3a. The driver requires a different bus due to capacity needs.	3a. The system allows operators to reassign buses based on route requirements.

Table 14. Bus Assignment and Vehicle Information

Use Case 15: Schedule Status Management

Use Case	Schedule Status Management
Actors	Bus Driver
Pre-condition	The driver is logged in and has active schedule assignments.
Post-Condition	The driver can update their schedule status and track completion.
Flow of Events	
Actor action	System Response
1. The driver views their current schedule status (scheduled, active, completed). 2. The driver updates their status when starting a route (changing from scheduled to active). 3. The driver marks routes as completed when finishing their assigned journey. 4. The driver reports any delays, cancellations, or incidents during their shift.	1. The system displays current schedule status and available status options. 2. The system updates the schedule status in real-time and notifies operators. 3. The system tracks completion times and calculates performance metrics. 4. The system logs incident reports and updates overall schedule tracking.
Alternative Scenarios	

Actor Action	System Response
1a. The driver needs to cancel a scheduled route due to an emergency. 2a. The driver experiences delays and needs to update estimated arrival times. 3a. The driver completes their shift early or requires overtime.	1a. The system allows emergency cancellation and notifies dispatch immediately. 2a. The system updates estimated times and alerts passengers and operators. 3a. The system adjusts schedule timing and calculates appropriate shift completion.

Table 15. Schedule Status Management

Use Case 16: Performance Tracking and Statistics

Use Case	Performance Tracking and Statistics
Actors	Bus Driver
Pre-condition	The driver has completed multiple schedules and has performance data.
Post-Condition	The driver can view their performance metrics and schedule history.
Flow of Events	
Actor action	System Response

1. The driver accesses their performance statistics from the profile section. 2. The driver reviews completion rates, total schedules, and performance metrics. 3. The driver checks their schedule history and recent assignments. 4. The driver uses performance data to improve their operational efficiency.	1. The system displays comprehensive performance statistics including completion rates. 2. The system shows total schedules, completed routes, active assignments, and cancellations. 3. The system provides historical schedule data with route details and timing information. 4. The system calculates performance trends and provides feedback for improvement.
Alternative Scenarios	
Actor Action	System Response
1a. The driver has limited performance history (new driver). 2a. The driver wants to compare their performance with other drivers. 3a. The driver needs to address performance issues or concerns.	1a. The system displays "Limited data available" and shows available metrics. 2a. The system provides general performance benchmarks while maintaining driver privacy. 3a. The system flags performance concerns and suggests improvement recommendations.

Table 16. Performance Tracking and Statistics

Use Case 17: Terminal Rental Management

Use Case	Terminal Rental Management
Actors	Terminal Manager
Pre-condition	Terminal manager is logged into the web platform. Bus operators have submitted rental applications.
Post-Condition	Applications are approved/rejected. Approved operators gain terminal access. Rejected operators are notified.
Flow of Events	
Actor action	System Response
1. The manager accesses the “Rental Applications” section. 2. The manager reviews application details and checks terminal availability. 3. The manager approves/rejects the application with comments.	1. The system displays pending applications with operator details, requested dates and spaces. 2. The system shows terminal occupancy and operator history. 3. The system updates application status:

	- Approved: Grants terminal access and schedules visibility. - Rejected: Remove access and notifies operator.
Alternative Scenarios	
Actor Action	System Response
1a. The manager requests application modifications.	1a. The system sends revision requests to the operator via in-app messaging.
2a. The operator submits an incomplete application.	2a. The system flags missing fields and block submission.

Table 17. Terminal Rental Management

Use Case 18: Terminal Space Allocation

Use Case	Terminal Space Allocation
Actors	Terminal Manager
Pre-condition	Rental application approved. Terminal layout data is available.
Post-Condition	Bays allocated to operators.

	Terminal layout updated with schedules.
Flow of Events	
Actor action	System Response
1. The manager opens the "Terminal Layout" visualization tool. 2. The manager drags approved bus schedules to specific bays. 3. The manager sets rental duration and saves.	1. The system displays an interactive terminal map with bay status (occupied/available). 2. The system validates assignments against schedule conflicts. 3. The system updates bay assignments, notifies operators, and syncs with schedules.
Alternative Scenarios	
Actor Action	System Response
1a. The manager assigns a bay under maintenance.	1a. The system blocks assignment and highlights maintenance schedule.

Table 18. Terminal Space Allocation

Use Case 19: Communication and Announcements

Use Case	Communication and Announcements
Actors	Terminal Manager

Pre-condition	Terminal is logged in. Bus operators are registered in the system.
Post-Condition	Alerts/messages delivered to bus operators. Announcements published.
Flow of Events	
Actor action	System Response
1. The manager selects “Messaging & Announcements” sections. 2. The manager drafts alerts/announcements (e.g., bay changes, disruptions.). 3. The manager schedules/publishes announcements.	1. The system shows templates, recipient filters (e.g., All North Terminal Bus Operators) 2. The system validates content and if any attachment support is included (PDFs or images). 3. The system pushes notifications to bus operators and logs delivery status.
Alternative Scenarios	
Actor Action	System Response
1a. Emergency alert (e.g., terminal closure).	1a. The system overrides “Do Not Disturb” settings and triggers SMS/Email fallback.

2a. Bus Operator replies to a message.	2a. The system threads conversations and notifies the terminal manager.
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Table 19. Communications and Announcements

Use Case 20: Schedule Visibility Access

Use Case	Schedule Visibility Access
Actors	Terminal Manager
Pre-condition	Terminal Manager is logged in. Bus operators have created schedules. Rental decisions are made.
Post-Condition	Approved operators' schedules become visible to the manager.
Flow of Events	
Actor action	System Response
1. The manager navigates to the Bus Schedules section. 2. The manager enters a date, selects a status, or chooses a driver to filter.	1. The system displays the bus schedules page with filter options (date, status, driver). 2. The system filters and displays the matching bus schedules.

3. The manager views details such as schedule ID, bus, driver, route, departure, arrival, status, and date.	3. The system ensures only approved and valid schedules are visible.
Alternative Scenarios	
Actor Action	System Response
1a. The manager applies filters but no schedules are available.	1a. The system displays a “No schedules found” message with a calendar icon.
2a. The manager does not apply any filters.	2a. The system displays all available approved schedules by default.

Table 20. Schedule Visibility Access

The Entity Relationship Diagram (ERD) depicts the relationship in the database, including attributes with their corresponding primary and foreign keys.

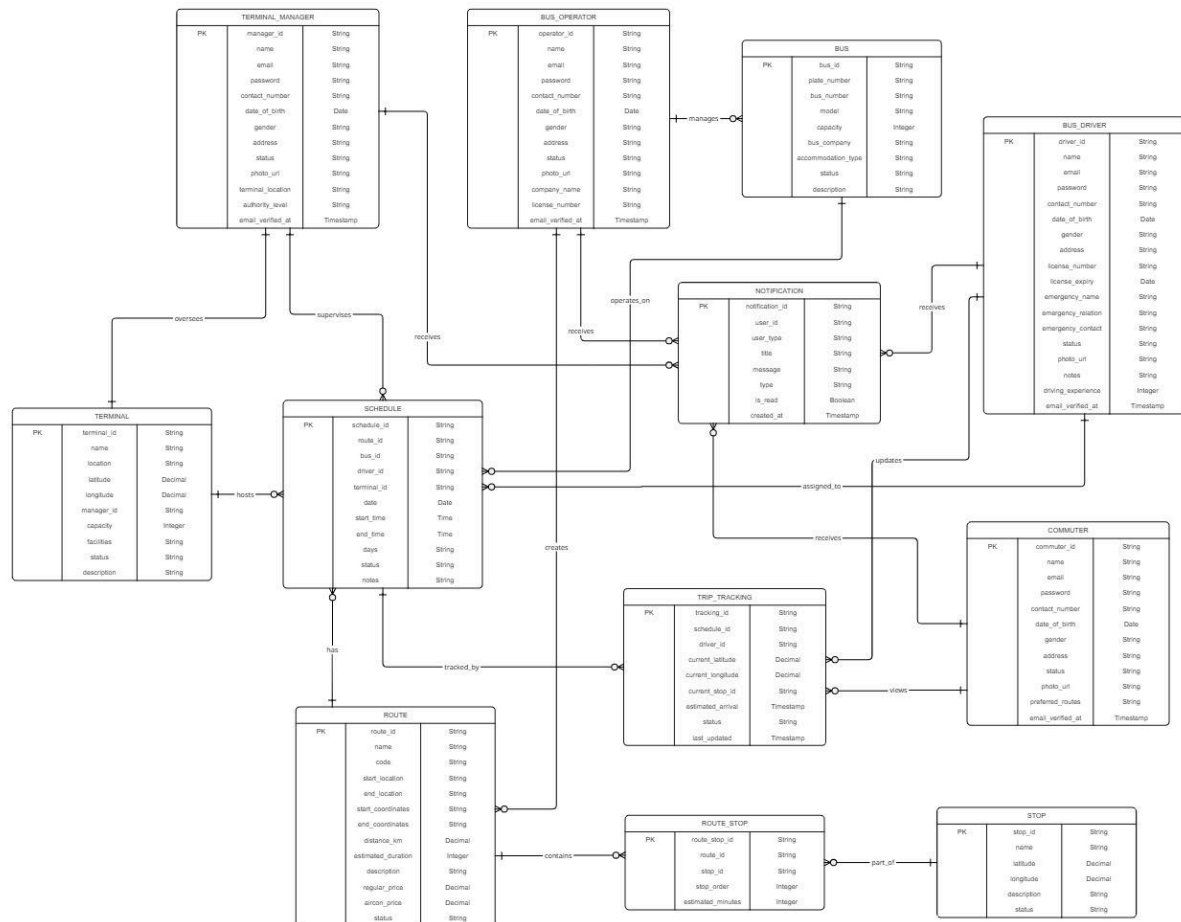


Figure 4.1 - Entity Relationship Diagram

Figure 4.1 illustrates the Entity Relationship Diagram (ERD) core structure of the TransiTrack bus management system. It includes key entities such as Bus Operators, Terminal Managers, Bus Drivers, Commuters, Buses, Routes, Schedules, Stops, Bookings, Trip Tracking, and Notifications. Each entity stores relevant information and connects through defined relationships. Bus Operators manage buses and routes, Terminal Managers oversee schedule operations and terminal stops, Bus Drivers are assigned to schedules and provide real-time trip tracking, and Commuters book trips and receive notifications. The diagram captures how data flows between the Laravel backend system and Firebase real-time database, supporting comprehensive bus

operation workflows, real-time tracking, and user interactions across web and mobile platforms.

CLASS DIAGRAM

The class diagram shows the system’s structure as classes with their attributes, operations, and relationships among other classes

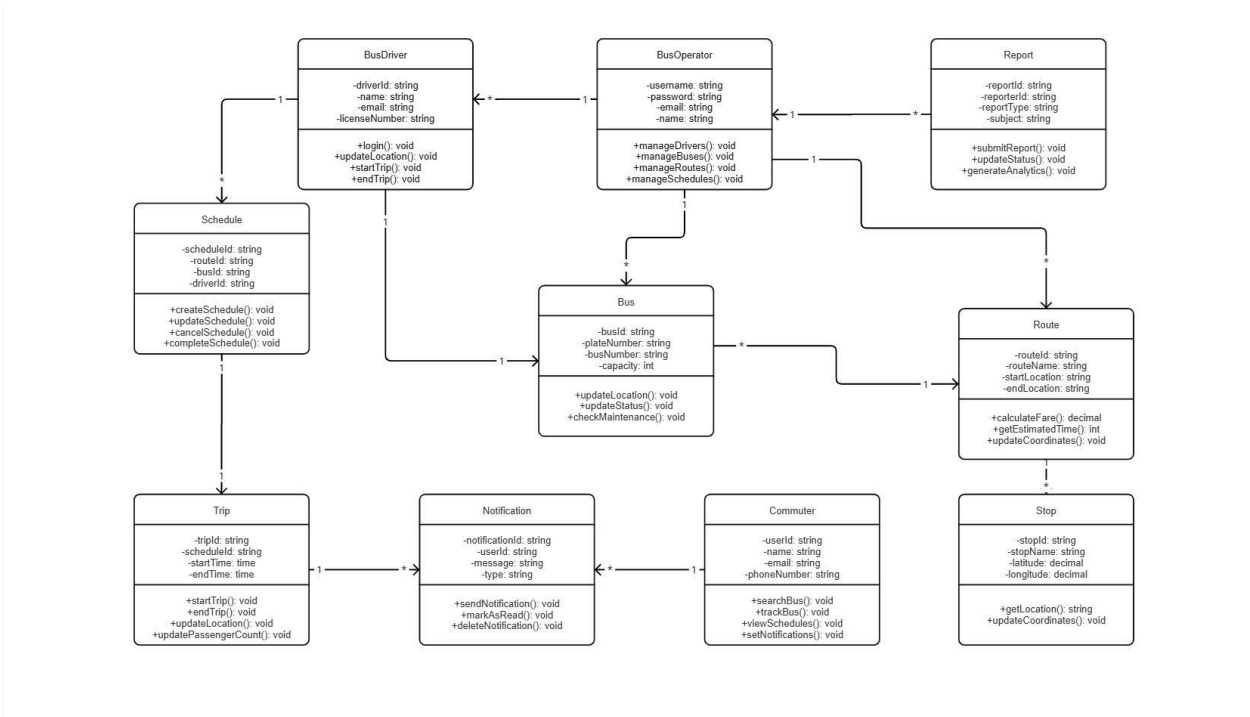


Figure 4.2 - Class Diagram

Figure 4.2 depicts the class diagram of the TransiTrack system, which outlines the key components and their relationships within the comprehensive bus transportation management platform. The diagram illustrates how the Bus Operator serves as the central management entity, overseeing drivers, buses, routes, and schedules while coordinating with Terminal Managers for efficient operations, where Bus Drivers are assigned to specific buses and follow designated routes while executing scheduled trips with capabilities for real-time location updates and trip status reporting, and Commuters interact with the system to track buses, view schedules, and receive notifications about their preferred routes. The Bus entity maintains essential vehicle information including capacity, status, and real-time location data, while Routes define the pathways between

locations with integrated fare calculation and timing estimates, and Stops represent physical locations along routes with precise GPS coordinates, with the Schedule entity acting as a crucial connector that links buses, drivers, and routes for specific time periods, which then generates individual Trip instances for real-time tracking and management. The system incorporates a comprehensive notification system that keeps all stakeholders informed of important updates, delays, and schedule changes, while the Report entity enables users to submit feedback, incidents, or service issues that are reviewed by Bus Operators for continuous service improvement, with all entities interconnected through one-to-many and one-to-one relationships that ensure seamless data flow, real-time tracking capabilities, and efficient coordination between all system components, ultimately providing a robust platform for modern bus transportation management.

Activity Diagram

An Activity Diagram is a type of UML diagram that illustrates the flow of actions or processes in a system.

1. User Login & Authentication

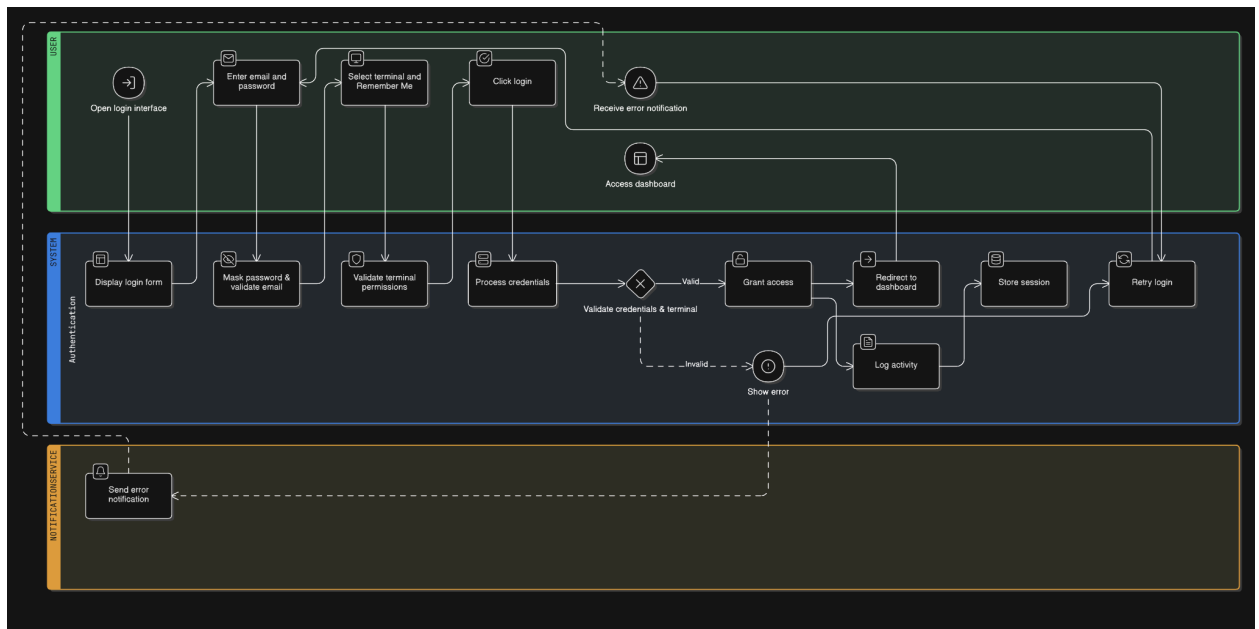


Figure 5.1 - User Login & Authentication

Figure 5.1 shows the login process, starting from when the user opens the login page and enters their email, password, and other options. The system then validates the inputs, checks terminal access, and processes the credentials. If valid, the user is granted access to the dashboard, a session is stored, and the activity is logged. If invalid, the system sends an error notification and allows the user to retry logging in.

2. Schedule Creation and Management

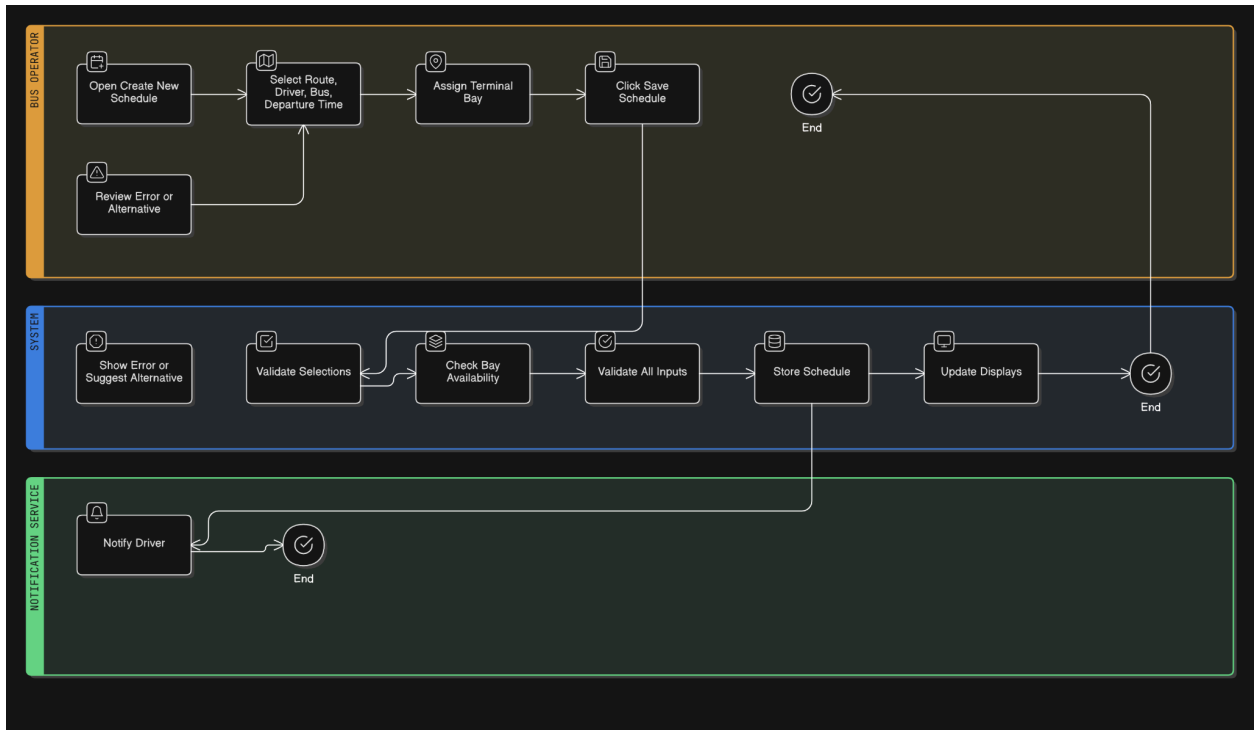


Figure 5.2 - Schedule Creation and Management

Figure 5.2 illustrates the schedule creation and management process for bus operations. The bus operator begins by creating a new schedule, selecting the route, driver, bus, and departure time, then assigning a terminal bay. The system validates the selections, checks bay availability, and verifies all inputs. If errors occur, the system suggests alternatives for the operator to review and adjust. Once inputs are valid, the schedule is stored, displays are updated, and the driver is notified. The process ends after successfully saving and communicating the schedule.

3. Monitoring & Driver Management

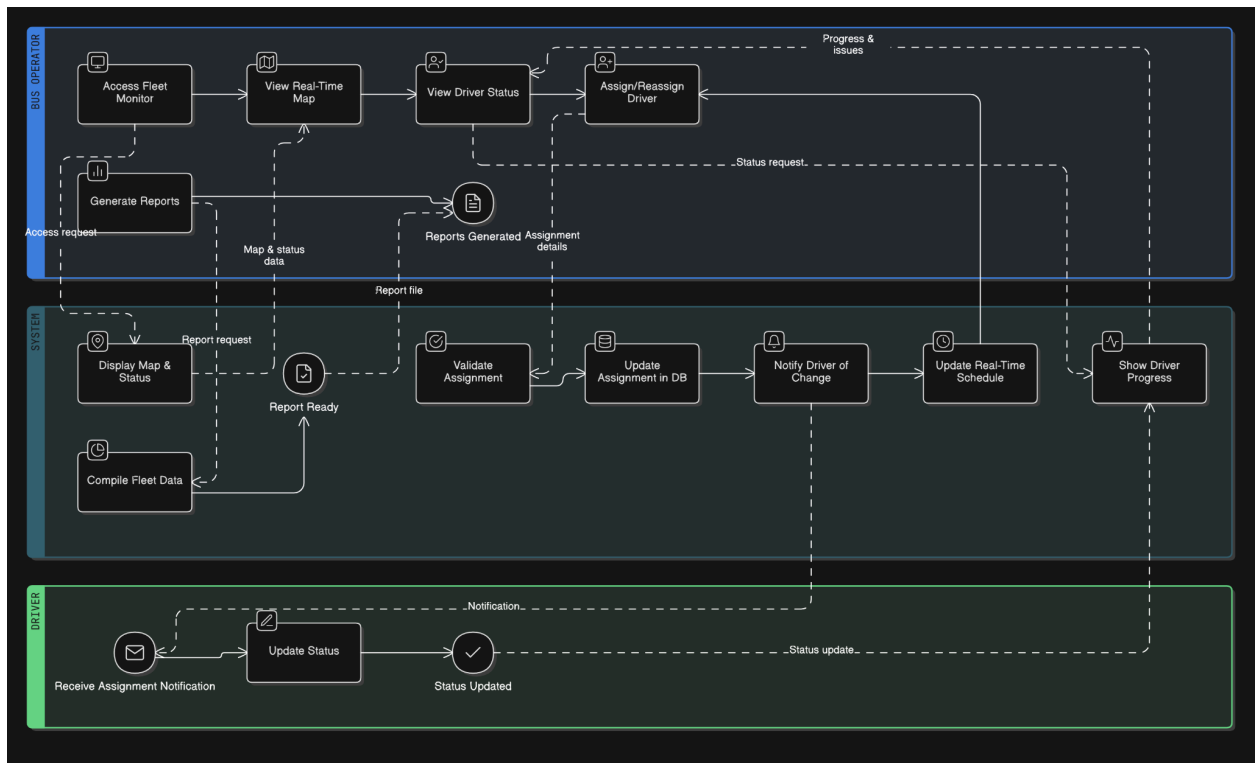


Figure 5.3 - Monitoring & Driver Management

Figure 5.3 illustrates the Monitoring & Driver Management process. The bus operator begins by accessing the fleet monitor, where they can view real-time maps, driver statuses, and assign or reassign drivers as needed. The system supports this by compiling fleet data, displaying maps and statuses, and validating assignments. Once validated, the system updates the assignment in the database, notifies the driver, and refreshes the real-time schedule. Meanwhile, drivers receive assignment notifications, update their status, and the system reflects these updates to show progress. Additionally, operators can generate reports that provide assignment details and status insights for monitoring.

4. Terminal Bay Management

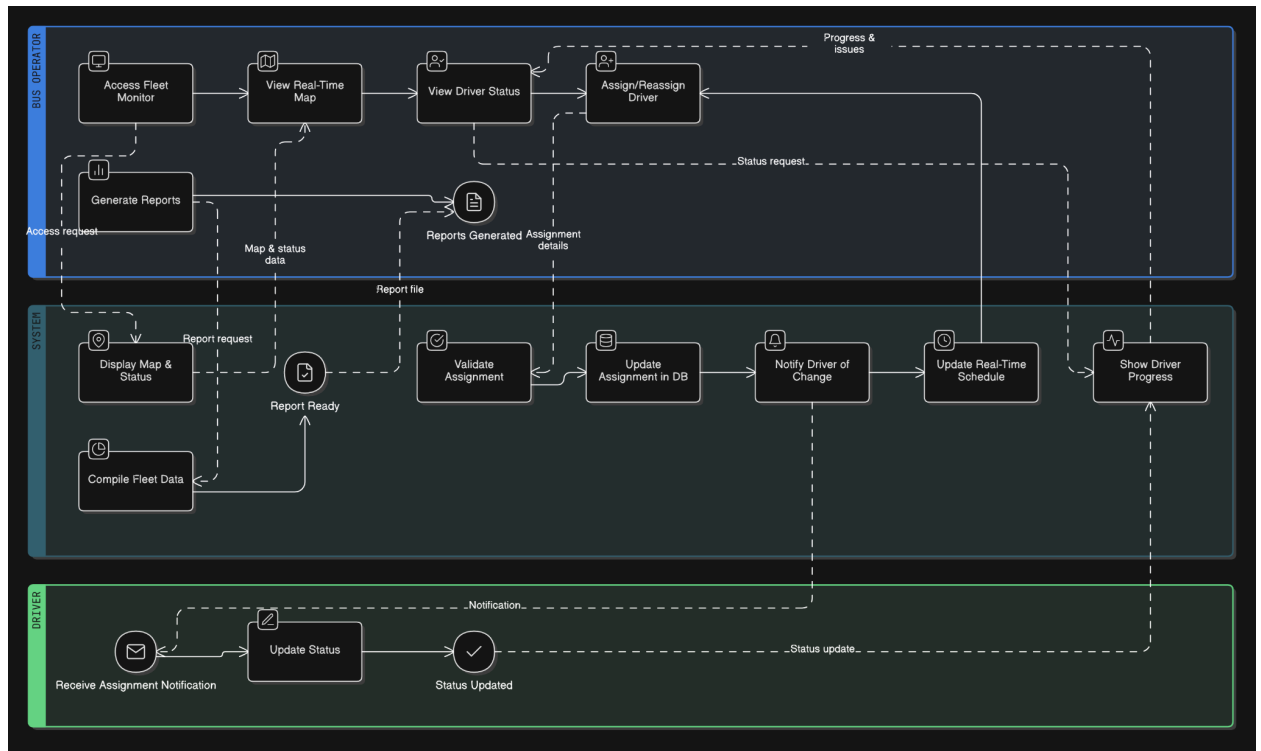


Figure 5.4 - Terminal Bay Management

Figure 5.4 shows the Terminal Bay Management process. The operator monitors fleet activity through real-time maps, driver statuses, and assignments while ensuring buses are properly allocated to terminal bays. The system compiles fleet data, validates assignments, updates the database, and reflects changes in the real-time schedule. Drivers are notified of updates and can report their status, which the system tracks to show progress. Reports are also generated to help operators manage bay usage and ensure efficient scheduling at the terminal.

5. Route Creation & Coordinate Management

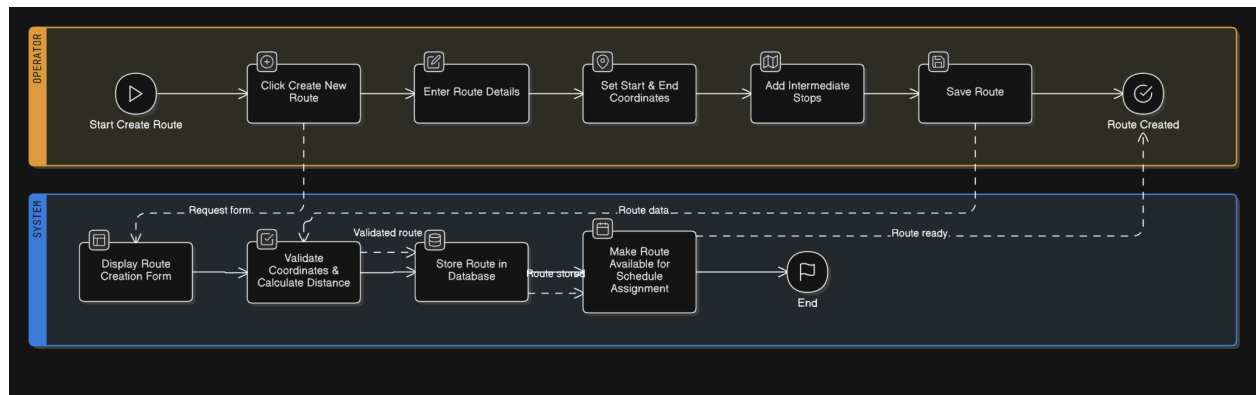


Figure 5.5 - Route Creation & Coordination Management

Figure 5.5 illustrates how a new bus route is created and managed within the system. The process begins when the operator initiates route creation by clicking Create New Route. The operator then enters the required route details, sets the start and end coordinates, and may add intermediate stops before clicking Save Route. On the system side, the application first displays the route creation form, then validates the coordinates and calculates the route distance. A decision point checks whether the entered route is valid. If valid, the route is stored in the database and made available for schedule assignment. If invalid, the system generates an error message and prompts the operator to review and correct the route details. The process ends successfully when the route is stored and marked as Route Created, making it ready to be used for scheduling and driver assignment.

6. Driver Assignment & Schedule Management

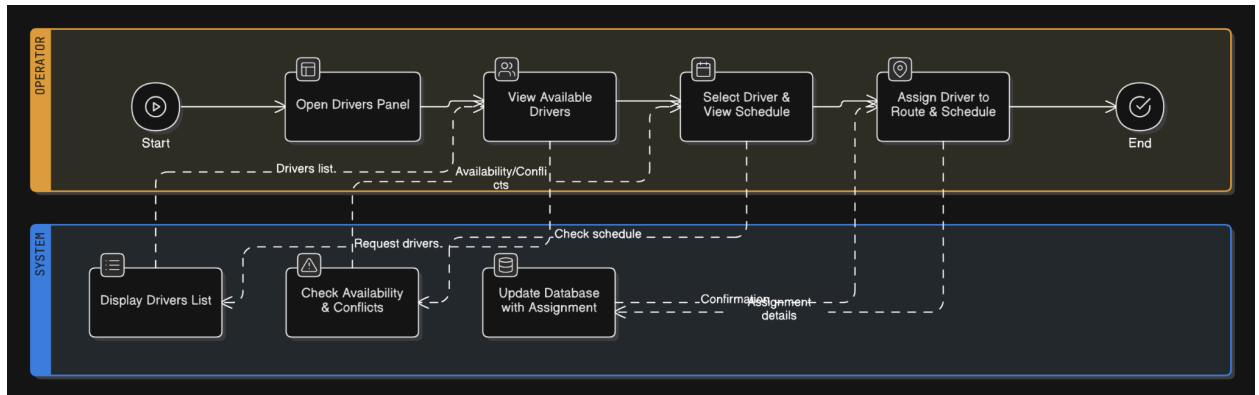


Figure 5.6 - Driver Assignment & Schedule Management

Figure 5.6 shows how drivers are assigned to routes and schedules. The operator opens the drivers panel, views available drivers, selects one, and assigns them to a route. The system displays the list, checks availability and conflicts, then updates the database with the new assignment. The process ends with the driver successfully scheduled for the chosen route.

7. Bus Fleet Management

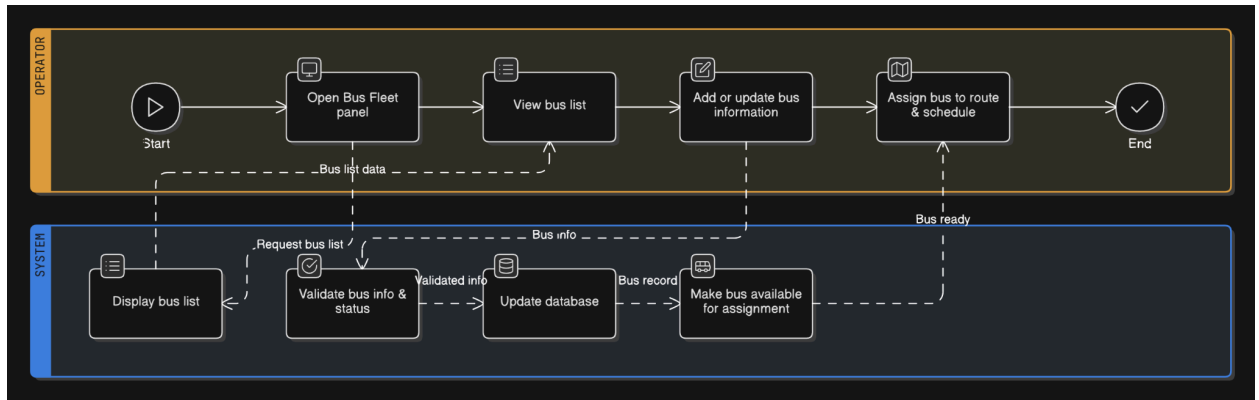


Figure 5.7 - Bus Fleet Management

Figure 5.7 illustrates how buses are managed in the system. The operator opens the bus fleet panel, views the list of buses, and adds or updates bus information. The system validates the details, updates the database, and makes the bus available for assignment. Finally, the operator assigns the bus to a route and schedule, completing the process.

USER INTERFACE DESIGN

Website Development

Register and Login (Bus Operator)

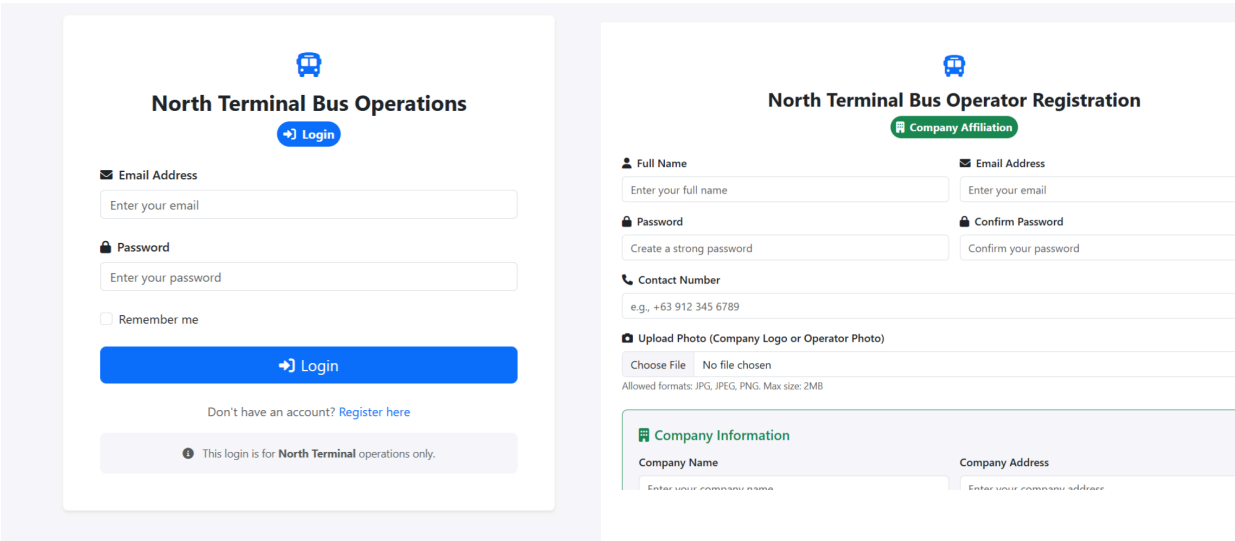


Figure 6.1 & Figure 6.2 - Register and Login

Figures 6.1 and 6.2 illustrate the login and registration interfaces for the North Terminal Bus Operator platform. The login view (Figure 5.1) provides a straightforward form where operators enter their email and password, with an option to remember their session for future logins. The design emphasizes clarity and accessibility, featuring visual cues and helpful feedback for invalid credentials. The registration view (Figure 5.2) guides new operators through a comprehensive sign-up process, collecting essential information such as contact details, company information, fleet size, and routes served. Users can also upload a profile photo to personalize their account. Both views are styled for ease of use, ensuring operators can quickly access or create their accounts and begin managing their operations efficiently.

Operator Panel (Bus Operator)

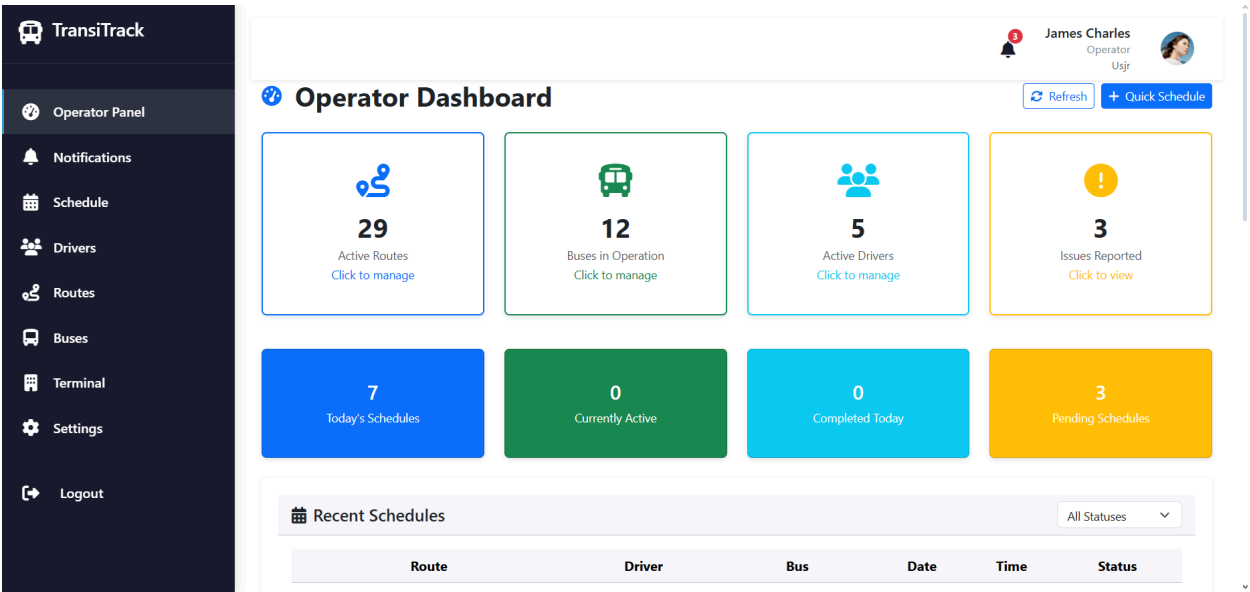


Figure 6.3 - Operator Panel

Figure 6.3 presents the Operator Panel for the Bus Operator system, serving as the central dashboard for managing daily operations at the North Terminal. The panel features a set of interactive cards that provide real-time statistics on active routes, buses in operation, drivers, and reported issues, each with quick access to relevant management pages. Below these, operators can view summarized schedule statistics for the day, including active, completed, and pending trips. The dashboard also displays a list of recent schedules with filtering options, enabling efficient tracking and management. Additional sections offer performance insights and quick alerts, while modals allow for detailed issue review and schedule management. The design prioritizes clarity, responsiveness, and ease of navigation, empowering operators to oversee and coordinate terminal activities effectively.

Schedule Management (Bus Operator)

TransiTrack

Operator Panel

Notifications

Schedule

Drivers

Routes

Buses

Terminal

Settings

Logout

James Charles

Operator

Ugjr

Create New Schedule

Schedule Management

Current Schedules

Filter by Driver

Filter by Route

Filter by Date

Filter by Status

All Drivers

All Routes

dd/mm/yyyy

All Statuses

Apply Filters

Clear Filters

Driver	Bus	Route	Date	Time	Fare	Status	Actions
Dexter Morgan	CS-002 Hino FB	Whatever Cebu North Bus Terminal (SM City) → 6002, Liloan, Cebu, Philippines	09/09/2025 Tuesday	02:24 AM to 08:24 AM	P50.00 A/C: P23.01	Scheduled	
Dexter Morgan	CR-001 Yutong ZK6119H	Cebu to Vistoria via Hagnaya Cebu North Bus Terminal → Vistoria	09/09/2025 Tuesday	03:55 PM to 06:55 PM	P140.00 A/C: P180.00	Scheduled	

Figure 6.4 - Schedule Management

Figure 6.4 showcases the Schedule Management panel for the Bus Operator system. This interface enables operators to efficiently create, assign, and oversee bus schedules within the terminal. At the top, users can quickly initiate new schedules by selecting routes, drivers, buses, and specifying trip details such as date and time. The form provides dynamic feedback, including route information and fare calculations, to assist in accurate scheduling. Below, the panel displays a comprehensive list of current schedules, complete with filtering options for driver, route, date, and status, allowing for targeted management and review. Operators can view, edit, or delete schedules directly from the table, with modals supporting detailed inspection and updates. The design emphasizes clarity and workflow efficiency, streamlining the process of coordinating daily bus operations.

Driver Management (Bus Operator)

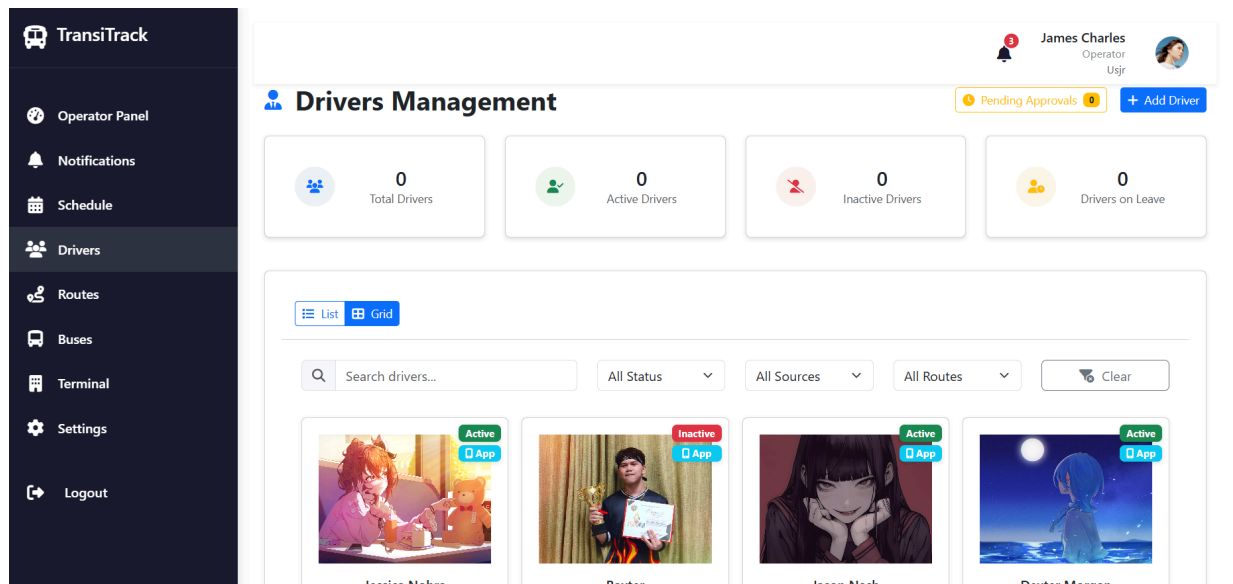


Figure 6.5 - Driver Management

Figure 6.5 illustrates the Driver Management panel within the Bus Operator system. This section provides operators with a comprehensive interface to oversee all registered drivers, track their status, and manage driver records efficiently. The dashboard displays key statistics such as total drivers, active, inactive, pending approvals, and those on leave. Operators can add new drivers using a detailed form that captures personal, license, and emergency contact information, with support for photo uploads. The panel offers both grid and table views for browsing driver profiles, along with search and filter options to quickly locate specific drivers or filter by status. Additional features include viewing pending registrations for approval, editing driver details, updating status, and accessing individual driver profiles. The design ensures streamlined management and oversight of driver personnel, supporting operational needs at the terminal.

Driver Profile (Bus Operator)

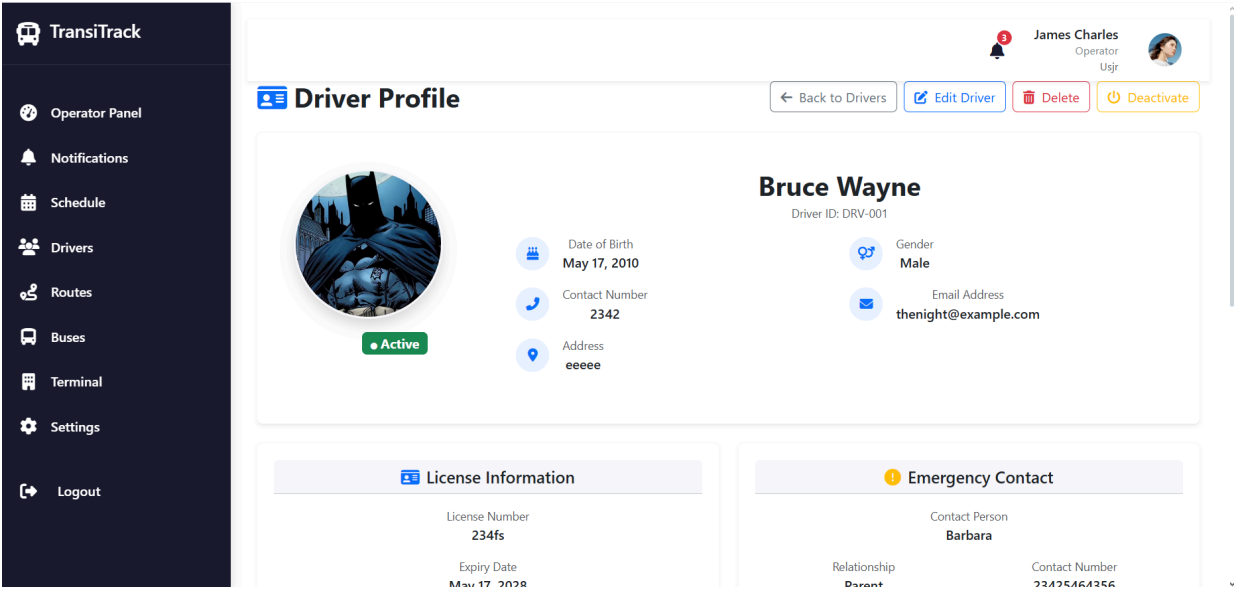


Figure 6.6 - Driver Profile

Figure 6.6 displays the Driver Profile view in the Bus Operator system. This panel provides a detailed overview of an individual driver’s information, including personal details, license data, emergency contacts, and a profile photo. Operators can quickly access actions to edit, delete, or toggle the driver’s active status. The profile card highlights the driver’s name, ID, and contact information, while additional sections present license validity and emergency contact details for safety and compliance. The interface also lists all assigned routes and schedules, allowing operators to monitor the driver’s operational assignments and history. If available, supplementary notes about the driver are shown for reference. The design ensures that all relevant driver data is easily accessible, supporting effective personnel management and operational oversight.

Route Management (Bus Operator)

TransiTrack

Operator Panel

Notifications

Schedule

Drivers

Routes

Buses

Terminal

Settings

Logout

James Charles

Operator

Usjr

+ Add Route

Route Management

Search Routes

Status

Search by name, code, or location...

All Status

Search

Clear

29 routes found

Code	Name	Start	End	Regular Fare	Aircon Fare	Duration	Distance	Status	Actions
RT-6002	Whatever	Cebu North Bus Terminal (SM City)	6002, Liloan, Cebu, Philippines	₱50.00	₱23.01	56 mins	18.70 km	Active	
ARGA-230	Argraaa	Cebu North Bus Terminal (SM City)	A. S. Fortuna Street, Casuntingan	₱20.25	₱26.32	10 mins	3.50 km	Active	
HEN-291	Henry The Pottah	Cebu North Bus Terminal (SM City)	Colon Street, Sambag II	₱20.85	₱27.11	12 mins	3.90 km	Active	

Figure 6.7 - Route Management

Figure 6.7 depicts the Route Management panel in the Bus Operator system. This interface allows operators to create, edit, and organize bus routes serving the North Terminal. The panel features a searchable and filterable list of all routes, displaying essential details such as route name, code, start and end locations, fare rates, and operational status. Operators can add new routes using an interactive form that includes map-based selection for destinations and stops, automatic route code generation, and fare calculation tools. The panel also supports viewing detailed route information, including assigned schedules and stops, through modals. The design streamlines the process of managing route data, ensuring accuracy and efficiency in planning and updating bus operations.

Bus Management (Bus Operator)

TransiTrack

Operator Panel
Notifications
Schedule
Drivers
Routes
Buses
Terminal
Settings
Logout

Bus Management + Add New Bus

Search by bus number, plate, or All Status All Types Search Clear

Bus Number	Plate Number	Model	Capacity	Company	Type	Status	Actions
JT-001	JLT-001	Hyundai County	45 seats	JULILA TRANSIT	Regular	Active	Edit Delete
JT-002	JLT-002	Hyundai County	45 seats	JULILA TRANSIT	Regular	Active	Edit Delete
IM-001	IMB-001	Hino RM2	50 seats	INDAY MEMIE BUS	Regular	Active	Edit Delete
CS-001	CSS-001	Hino FB	48 seats	CEBU SAN SEBASTIAN LINER CORP.	Regular	Active	Edit Delete
CS-002	CSS-002	Hino FB	48 seats	CEBU SAN SEBASTIAN LINER CORP.	Regular	Active	Edit Delete
RR-001	RR-001	Mitsubishi Rosa	40 seats	ROUGH RIDERS / WHITE STALLION	Regular	Active	Edit Delete
MC-001	MCA-001	Mitsubishi Fuso	46 seats	METRO CEBU AUTOBUS	Regular	Active	Edit Delete

Figure 6.8 - Bus Management

Figure 6.8 illustrates the Bus Management panel in the Bus Operator system. This section enables operators to oversee the fleet by adding, editing, and organizing bus records. The interface provides a searchable and filterable table of all buses, displaying details such as bus number, plate number, model, capacity, company, accommodation type, and operational status. Operators can quickly add new buses or update existing ones using a modal form that supports validation and error feedback. The panel also allows for filtering by status and accommodation type, making it easy to locate specific buses or review those in maintenance or inactive. The design supports efficient fleet management, ensuring that bus data is accurate and up-to-date for daily operations.

Terminal Layout (Bus Operator)

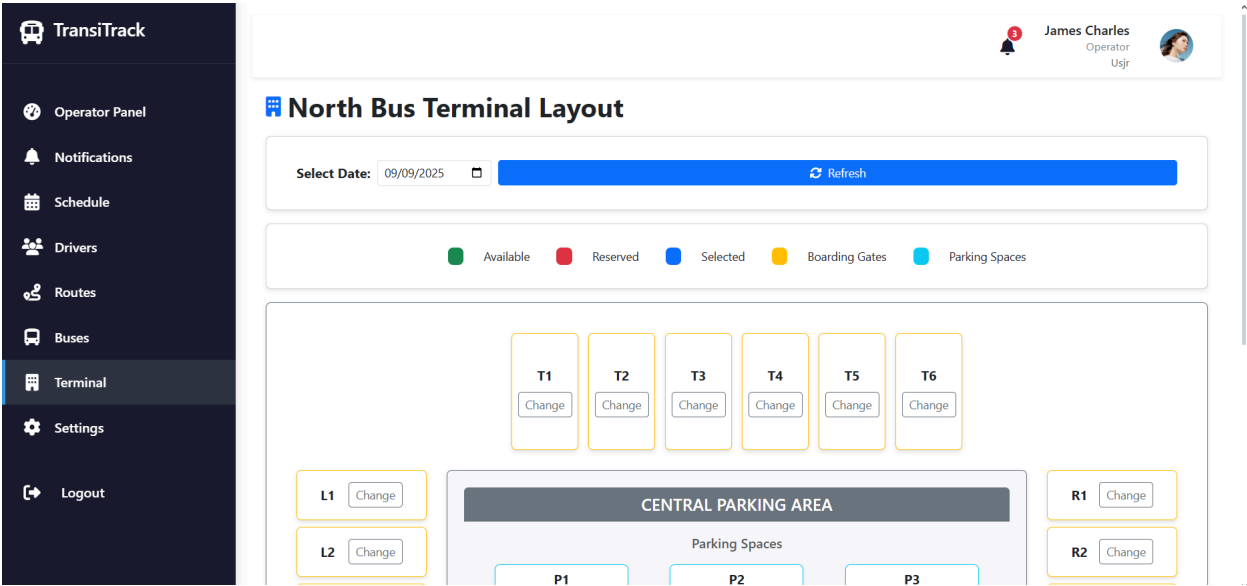
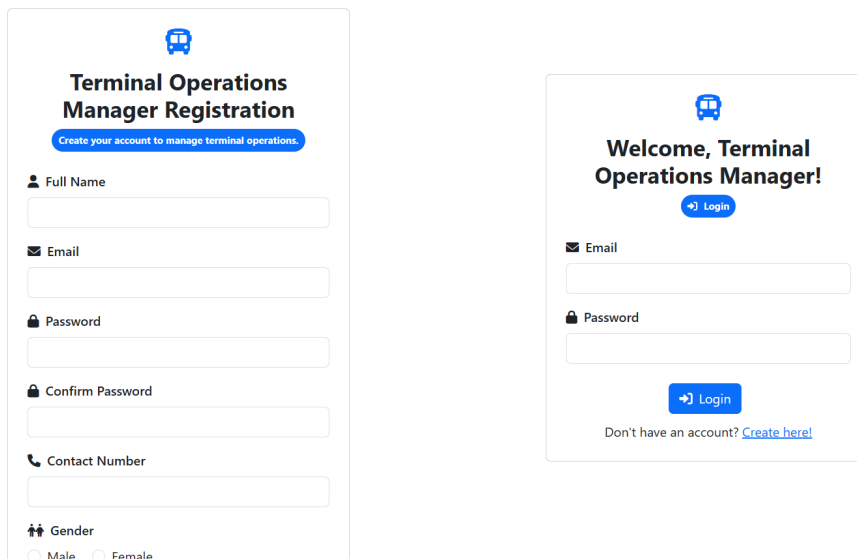


Figure 6.9 - Terminal Layout

Figure 6.9 presents the Terminal Management panel in the Bus Operator system, designed to visualize and coordinate activities within the terminal. The interface displays a detailed layout of boarding gates and parking spaces, with color-coded indicators for availability, reservations, and space types. Operators can select a date, view current bookings, and refresh the layout for real-time updates. The panel supports interactive actions such as reserving gates for specific drivers and buses, setting time limits, and changing assigned routes. Modals facilitate quick reservation and route assignment, while a summary table tracks all active and past bookings. The design streamlines terminal operations, ensuring efficient allocation of spaces and smooth coordination of bus movements throughout the day.

Register and Login (Terminal Manager)



The image displays two side-by-side screenshots of the Terminal Operations Manager web interface. The left screenshot shows the registration form, titled "Terminal Operations Manager Registration", with a blue button "Create your account to manage terminal operations". The form includes fields for Full Name, Email, Password, Confirm Password, Contact Number, and Gender (Male/Female). The right screenshot shows the login form, titled "Welcome, Terminal Operations Manager!", with a blue button "Login". It includes fields for Email and Password, and a link "Don't have an account? [Create here!](#)".

Figure 7.1 & Figure 7.2 - Register and Login

Figures 7.1 and 7.2 present the login and registration interfaces for the Terminal Operations Manager platform. The login interface (Figure 7.1) offers a simple and direct form where managers can securely enter their email and password to access the system, with a link to create an account if they are new users. The layout highlights clarity and professionalism, using recognizable icons and a clean design for ease of navigation. The registration interface (Figure 7.2) provides a structured form for new managers to create an account. It collects essential details including full name, email, password (with confirmation), contact number, gender, and upload photo for the user's profile. This ensures that sufficient user information is captured to support account security and personalization. Both interfaces are designed with a consistent style, emphasizing usability and accessibility, so managers can quickly log in or register and begin overseeing terminal operations efficiently.

Manager Panel (Terminal Manager)

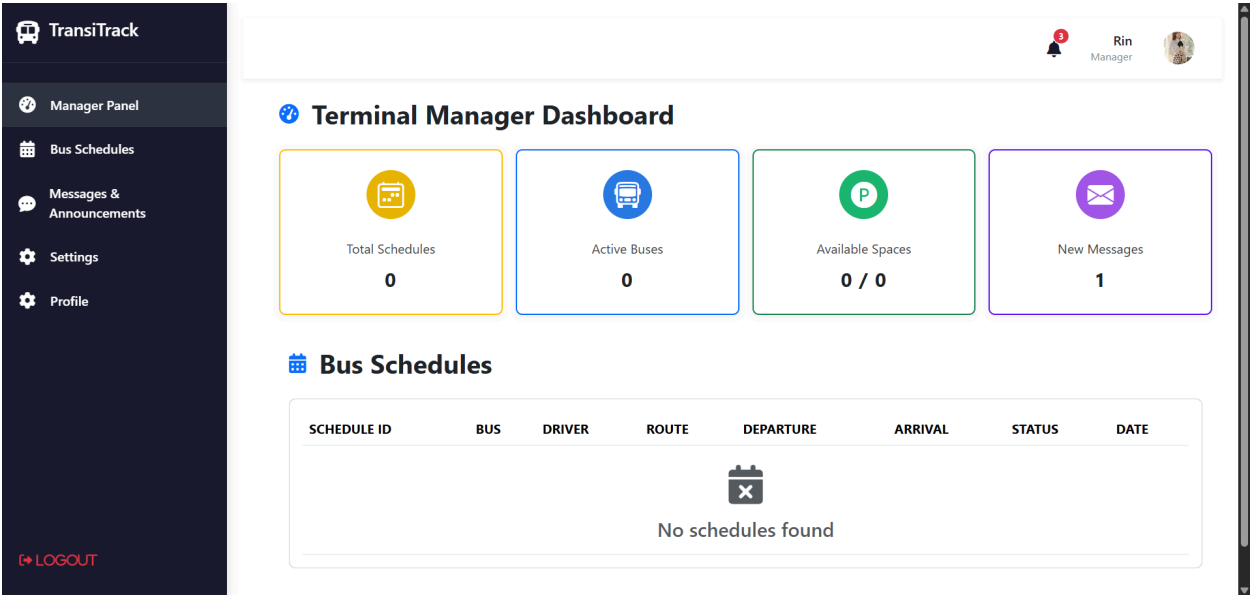


Figure 7.3 - Manager Panel

Figure 7.3 illustrates the Manager Panel dashboard of the Terminal Operations Manager system. This interface serves as the central hub where terminal managers can oversee and manage operations efficiently. The dashboard provides a quick overview through key indicators, including the total number of schedules, active buses, available parking spaces, and new messages. A dedicated section for bus schedules allows managers to track schedule IDs, assigned buses, drivers, routes, departure and arrival times, status, and dates.

Bus Schedules (Terminal Manager)

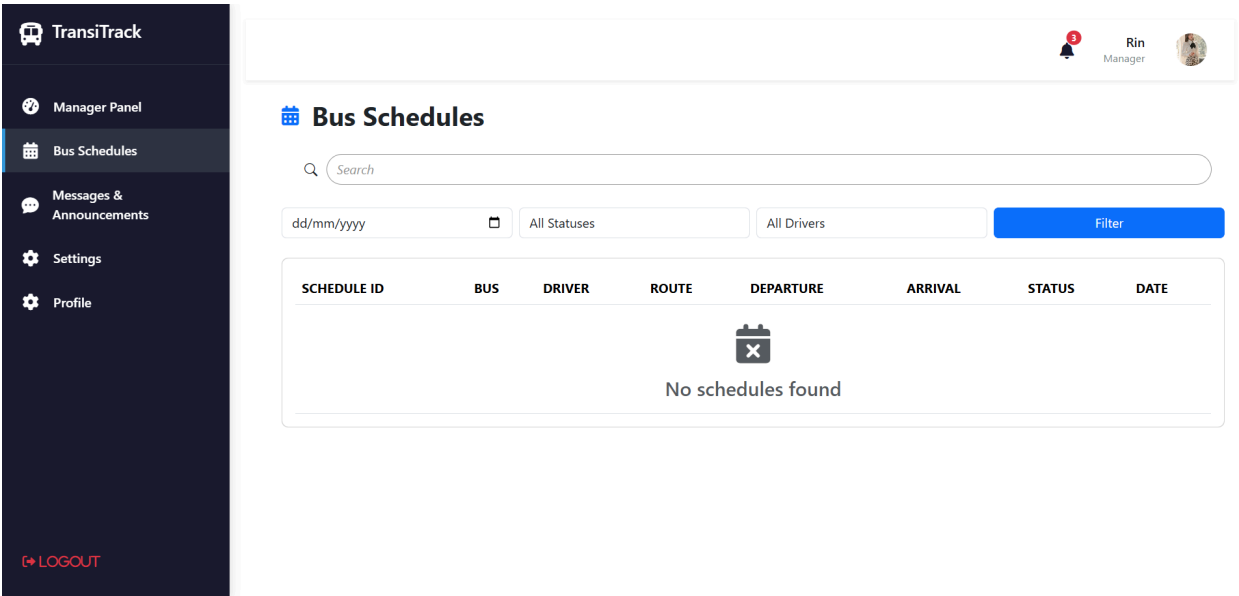


Figure 7.4 - Bus Schedules

Figure 7.4 shows the Bus Schedules interface within the Terminal Operations Manager system. This module enables managers to view and organize bus schedules in a structured table format, displaying details such as schedule ID, bus, driver, route, departure, arrival, status, and date. The interface includes filtering options, allowing schedules to be searched by date, status, or driver, making it easier to locate specific entries. When no schedules are available, the system displays a clear “No schedules found” message, ensuring users are informed about the absence of data.

Messages and Announcements (Terminal Manager)

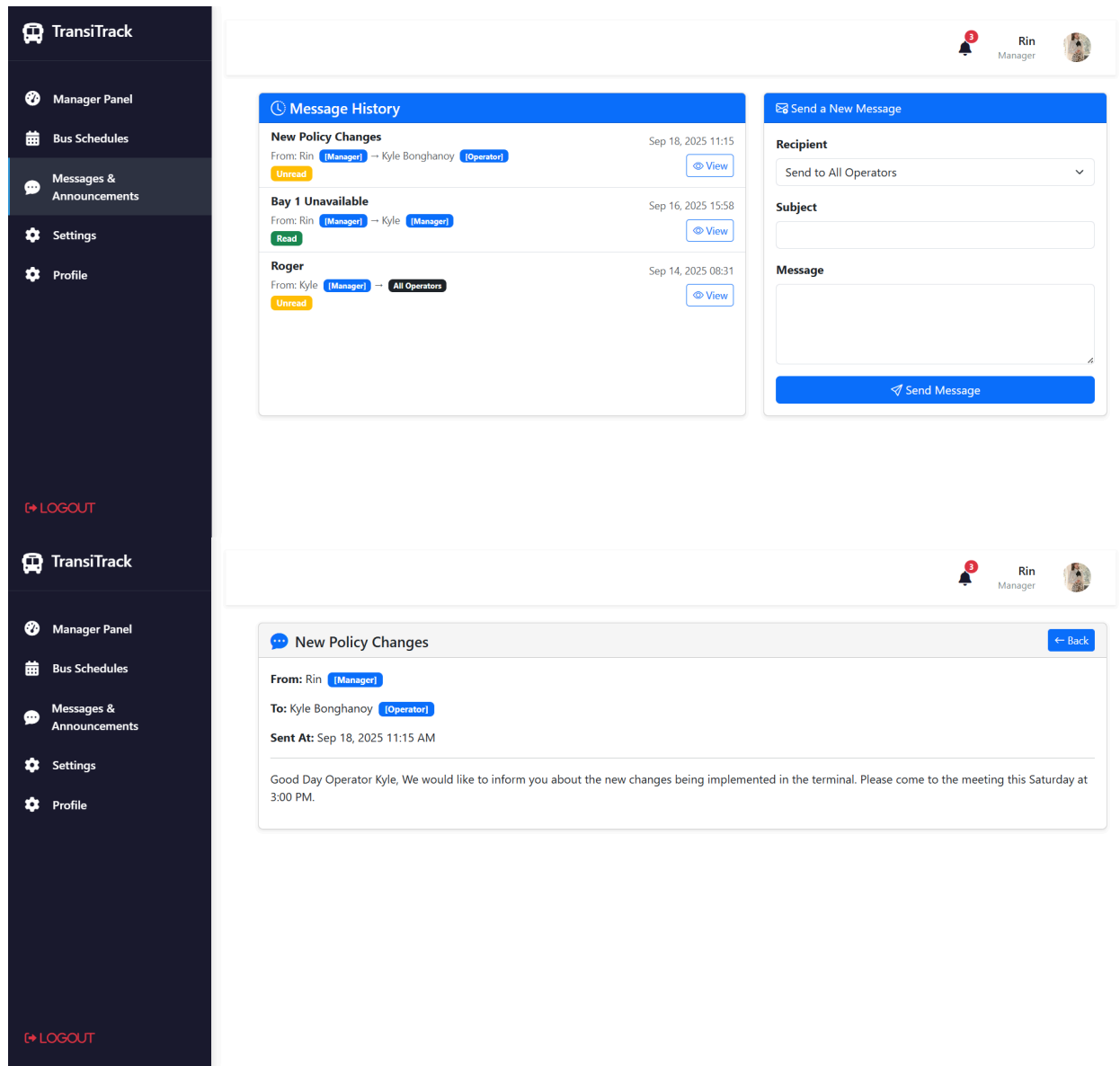


Figure 7.5 & Figure 7.6 - Messages and Announcements

Figure 7.5 illustrates the Messages & Announcements module where managers can compose and send new messages. The form allows the manager to specify the recipient role (e.g., all operators), add a subject line, and write the message body before sending. This feature facilitates clear and direct communication within the terminal, making it easier to distribute announcements, policy updates, and other important information to the intended recipients. Figure 7.6 displays the message viewing interface. This module allows users to open and review messages in detail, showing essential metadata such as the sender, recipient role, and timestamp.

Mobile Development

Login and Registration (Bus Driver)

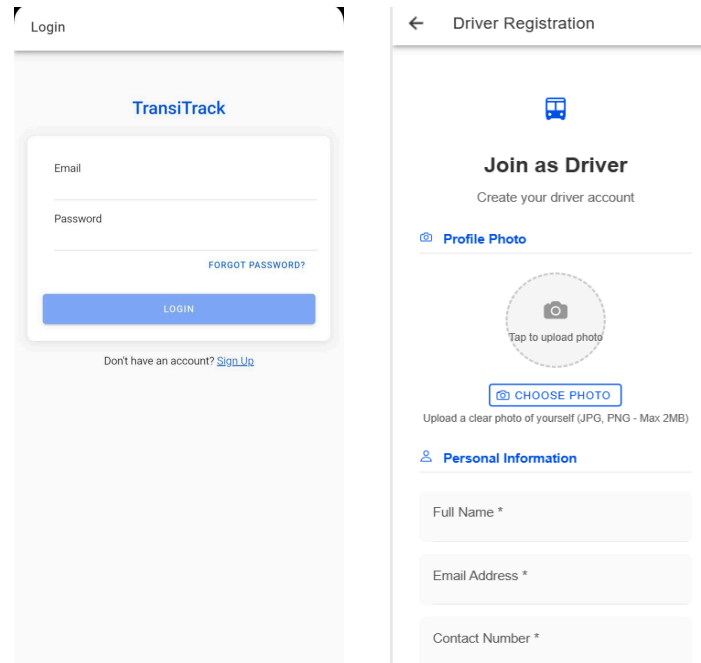


Figure 8.1 & Figure 8.2 - Login Page and Registration Page

Figures 8.1 and 8.2 illustrate the login and registration screens for Bus Drivers in the mobile application. The login view (Figure 8.1) offers a simple form for drivers to enter their email and password, enabling quick and secure access to their accounts. The registration view (Figure 8.2) guides new drivers through a detailed sign-up process, collecting essential information such as personal details, contact information, license data, emergency contacts, and a profile photo. The form includes validation and user-friendly feedback to ensure accuracy. Both screens are designed for clarity and ease of use, allowing drivers to efficiently authenticate or create their accounts and begin managing their schedules and assignments.

Home Page (Bus Driver)

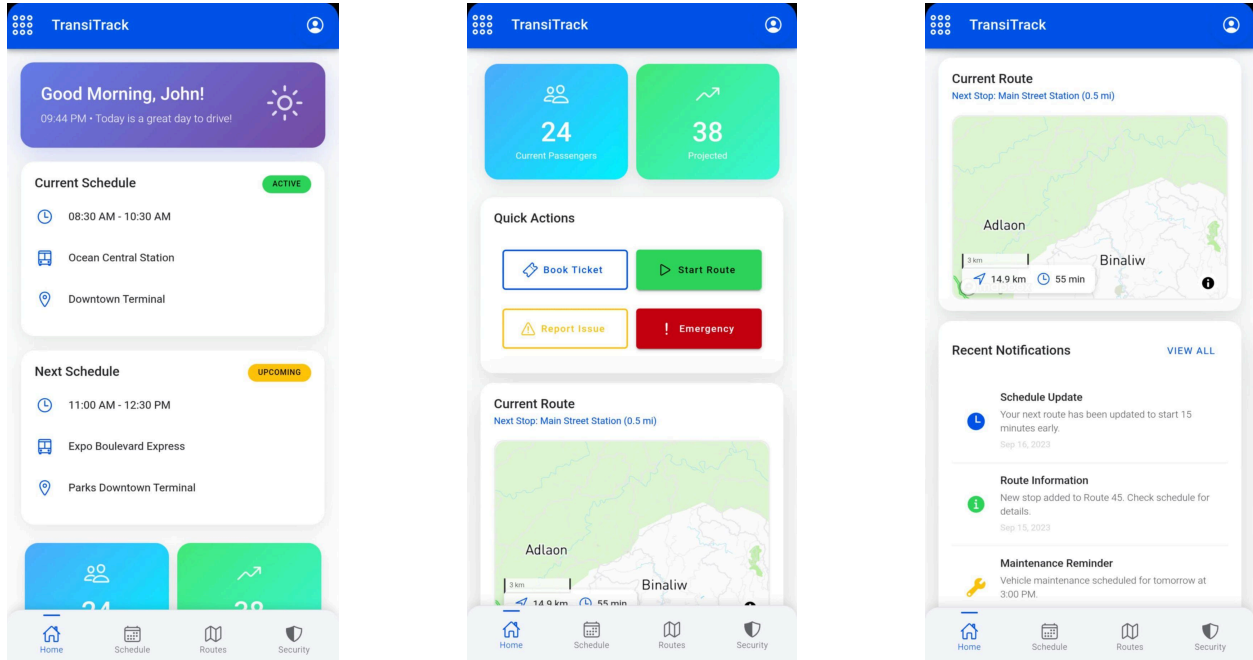


Figure 8.3 - Home Page

Figure 8.3 displays the Home page of the Bus Driver mobile application. This screen serves as the central hub for drivers, providing quick access to essential features and information. Upon logging in, drivers are greeted with a summary of their upcoming schedules, recent trips, and important notifications. The layout is designed for clarity and ease of navigation, allowing users to view their assignments, check route details, and access support resources. Interactive elements and visual highlights ensure that drivers can efficiently manage their daily tasks and stay informed about updates or changes in their operations.

Schedule Page (Bus Driver)

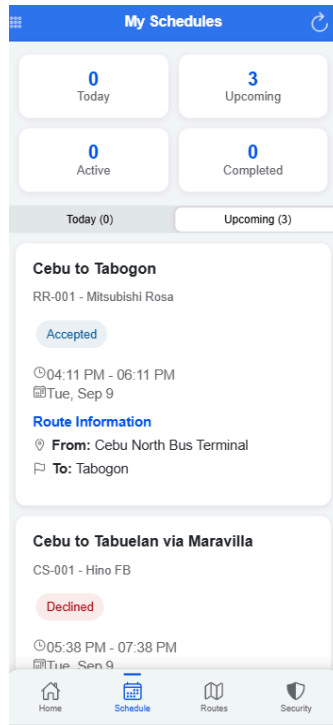


Figure 8.4 - Schedule Page

Figure 8.4 presents the Schedule page of the Bus Driver mobile application. This screen is dedicated to helping drivers manage and review their assigned trips efficiently. Upon accessing the page, drivers are shown a segmented view of their schedules, including today's assignments, upcoming trips, and completed journeys. Each schedule entry displays key details such as route name, bus information, departure and arrival times, fare rates, and status indicators. Interactive actions allow drivers to accept, decline, start, or complete schedules directly from the interface. The design emphasizes clarity, with color-coded status tags and organized lists, ensuring drivers can quickly identify their tasks and respond to changes. The Schedule page streamlines daily operations, keeping drivers informed and in control of their work assignments.

Routes Page (Bus Driver)

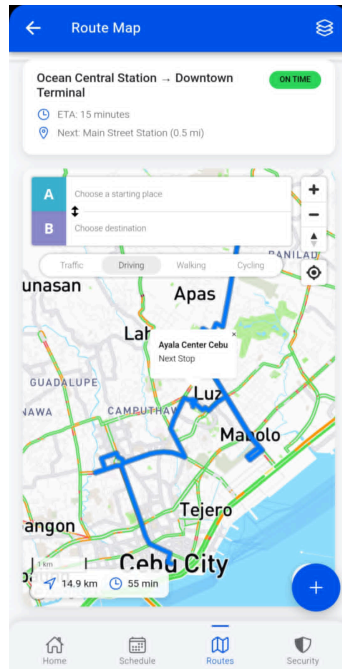


Figure 8.5 - Route Page

Figure 8.5 illustrates the Route page of the Bus Driver mobile application. This screen becomes accessible after a driver accepts an assigned schedule, providing a detailed view of the current route and a list of all accepted schedules. The top section highlights the active route, displaying essential information such as start and end locations, waypoints, estimated travel time, and bus details. An interactive map component visually traces the route, helping drivers navigate efficiently. Below, a schedule list presents all accepted assignments, allowing drivers to review upcoming trips and switch between routes as needed. The Route page is designed for clarity and real-time guidance, ensuring drivers have all necessary details to complete their journeys safely and on time.

Security Page (Bus Driver)

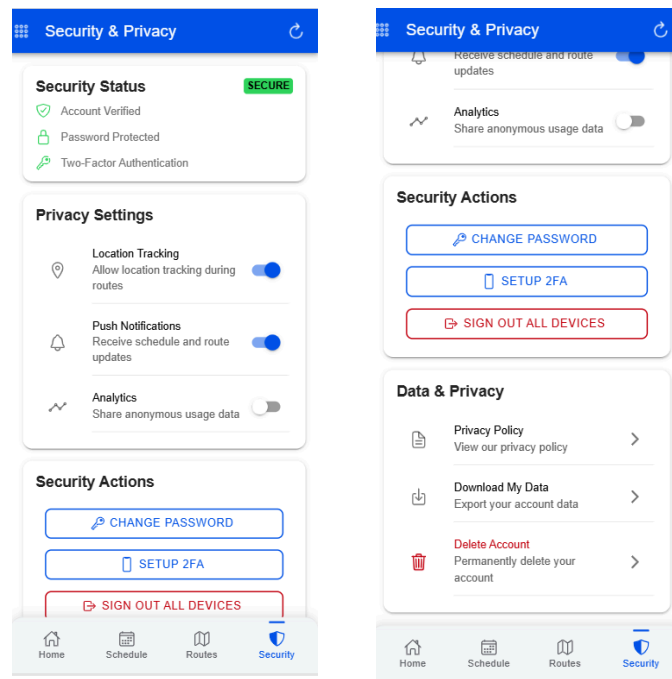


Figure 8.6 - Route Page

Figure 8.6 shows the Security page of the Bus Driver mobile application. This screen is designed to help drivers manage safety-related tasks and access important security resources during their operations. The page provides quick links to emergency contacts, incident reporting forms, and safety guidelines issued by the bus operator. Drivers can easily submit security alerts or request assistance if needed, ensuring rapid response in case of emergencies. The layout prioritizes clarity and accessibility, with prominent buttons and instructions for urgent actions. By centralizing security features, the Security page empowers drivers to maintain a safe environment for themselves and their passengers throughout their routes.

